

Avinash Karri

AI Engineer — Production AI, Multi-Agent Systems & ML Platforms

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Summary

AI Engineer with hands-on experience building and deploying **production-grade AI systems** spanning automation, multi-agent orchestration, semantic search, and ML inference. Strong background in **Python engineering, FastAPI services, vector retrieval, and AI observability**, with a track record of delivering measurable efficiency gains in enterprise environments. Known for a problem-solver mindset, I enjoy tackling ambiguous, hard problems and turning seemingly impossible challenges into scalable, reliable solutions through disciplined engineering and experimentation.

Technical Skills

AI/ML & GenAI: LLMs, RAG, Transformers, LangChain, LangGraph, Prompt Engineering, Embeddings, MCP

ML Frameworks: Scikit-learn, PyTorch (basic), TensorFlow (basic), Azure OpenAI, Vertex AI

Programming: Python, SQL, JavaScript, ReactJS

Backend & APIs: FastAPI, REST API Design, GitHub API, GCP APIs, LLM APIs

Cloud & Infra: Azure, GCP, Docker, CI/CD, Kubernetes

Data Engineering: ETL/ELT, Pipelines, Data Modeling, Vector Databases (Chroma)

Tools & Databases: Git, Jupyter, PostgreSQL, MongoDB

Work Experience

ALSTOM, Bengaluru

Jul 2024 – Present

Associate System Application Design Engineer

- Designed and deployed Python-based AI and automation systems eliminating **100% manual intervention** across recurring engineering workflows.
- Built and productionized a **RAG-based Semantic Document Intelligence platform** using embeddings and vector search, reducing document lookup time by **65%** and improving delivery efficiency by **20–25%**.
- Developed **FastAPI-based AI services** for document processing, data extraction, and semantic search, supporting multiple internal teams with standardized APIs.
- Collaborated with cross-functional engineering and domain teams to translate business requirements into deployable AI solutions following enterprise SDLC and controlled rollout practices.

Project Experience

Project Cypher — AI Agent Platform & MCP Integration

- **Tech stack:** Python, GenAI, LLMs, MCP, Agentic AI, ReactJS, GCP
- Built a **multi-agent orchestration system** enabling autonomous task execution across system, productivity, and automation workflows with contextual memory.
- Implemented **MCP-based structured context injection** to ground agent decisions using documents, APIs, and repository metadata.
- Developed an **observability dashboard** tracking agent execution traces, tool usage, latency, and failures, enabling **70% faster root-cause analysis**.

- Introduced performance metrics (task duration, success rate, retries) achieving **40% faster task execution** over baseline runs.

Asteria AIOps — Production ML & GenAI Platform

- **Tech stack:** Python, FastAPI, TorchServe, Docker, LLMs, ChromaDB, GitHub Actions, Azure/GCP
- Built a full-stack AIOps platform combining traditional ML and LLM-based anomaly detection with semantic log intelligence.
- Engineered RAG pipelines using ChromaDB for root-cause analysis and structured reasoning over logs and metrics.
- Implemented CI/CD workflows, containerized deployments, telemetry, and alerting, reducing investigation time by **45%**.

Enterprise RAG Chatbot — Document Intelligence

- **Tech stack:** Python, FastAPI, LLMs, Azure, Azure OpenAI, Streamlit
- Built a secure, multi-user semantic search platform for large enterprise document repositories using hybrid full-text and vector retrieval.
- Implemented contextual search with relevance ranking and highlighted matches, improving retrieval efficiency by **40%**.

AI DevOps Tool — LLM-Driven Log Analyzer

- **Tech stack:** Python, FastAPI, Mistral, Groq, Streamlit
- Implemented LangChain-based prompt orchestration and agent workflows for automated log analysis and recommendations.
- Built semantic log intelligence using embeddings and RAG pipelines for anomaly reasoning and diagnostics.

Certifications & Learning

Google Project Management Professional Certificate

https://www.credly.com/badges/5190d42b-e260-4612-93ca-1bb353bb19cc/linkedin_profile

Google Cloud Skills Profile

https://www.skills.google/public_profiles/3d9b435f-db59-420c-9b2f-963f28975d5e

Microsoft Learn Profile

<https://learn.microsoft.com/en-us/users/avinashkarri-0414/>

IBM RAG and Agentic AI

Education

GITAM University, Visakhapatnam

Bachelor of Technology, Computer Science Engineering

Apr 2024